Numerical Study of an Industrial Scale Continuous Container Glass Annealing Furnace

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Abstract. Annealing is a critical heat treatment process in container glass manufacturing, used to relieve residual stresses caused by rapid cooling during forming. This process takes place in continuous annealing furnaces, where glass is initially heated to the annealing temperature and then cooled at a carefully controlled rate to avoid any new stress development. Continuous annealing furnaces consist of temperature-controlled zones and the heat transfer rate to glassware is usually regulated with the set value of the zones. However, controlling the heat treatment process solely based on the zone temperature could be misleading. In addition to the zone air temperature, the operational condition of the tubular burner might also affect the heat transfer in the furnace.

This study presents a numerical investigation of the annealing process in an industrial-scale continuous container glass annealing furnace. A detailed conjugate thermal model of the furnace is developed where all the heat transfer modes inside the furnace were considered. Special attention was given to the modeling of the radiation heat transfer where the glassware was considered in full-geometrical detail. The model was validated with the furnace temperature data. Next, two case studies were conducted to study the effect of changing the position of the active tubular burners inside the heating zone of the furnace. Results have shown that changing the active burner positions does have a rather weak effect on the air temperature inside the furnace but a more considerable effect on the glassware temperatures. Also, activating the tubular burners away from the furnace inlet and deactivating the burners at the furnace entry results in a more gradual and uniform heating pattern on glassware. This study signifies the importance of active burner positions on the heat transfer in the furnace.

Keywords: Continuous Annealing furnace, Heat treatment, Numerical simulation, Conjugate Heat Transfer

1 Introduction

Annealing is a vital step in container glass manufacturing, relieving residual stresses formed during rapid cooling in the forming stage. If unaddressed, these stresses can lead to warping, fractures, or optical defects during product use. As emphasized by [1,5,6], effective stress relief depends on a precisely controlled thermal cycle especially during the transformation range of the cooling phase, where structural relaxation

governs stress evolution. Despite its importance, industrial annealing processes are often managed through trial-and-error adjustments of zone temperatures, conveyor speeds, and fan settings. This empirical approach risks compromising thermal uniformity and product quality, particularly in high-throughput settings. Numerical modeling offers a cost-effective, physics-based alternative to optimize furnace operation without disrupting production.

Recent studies have introduced various numerical techniques to simulate industrial annealing processes with greater accuracy. Capturing the full thermal behavior of these systems requires conjugate heat transfer solvers, as conduction and radiation interact strongly within furnace enclosures. Yıldız et al. [2] developed a segregated solver that decouples solid conduction and surface-to-surface radiation, allowing greater flexibility in implementation. The results indicated that Radiative heat fluxes were more dominant overall than Convective heat fluxes. Altun et al. [3] focused on the radiative heating performance of a continuous glass annealing furnace and demonstrated that container spacing and geometry-induced shadowing have a direct influence on temperature uniformity. Their findings emphasize that improper spacing can lead to thermal gradients and reduced product quality. Dillon [4] proposed a dual-solver CFD approach for transient simulations, where solid and fluid domains are solved separately. This framework enhances computational scalability; the results showed an agreement with the experimental measurements on the bottom of the furnace. On the operational side, Hajaliakbari and Hassanpour [7] performed a detailed energy performance analysis of a continuous annealing furnace for steel processing. One interesting finding was that burners of heating zone was less considerable than the heating power provided from the first or second preheating, contributing to both efficiency and product consistency. In annealing applications, the use of conjugate solvers incorporating various radiative models has become increasingly common. Earlier examples [8,9,10] applied finite-element, radiosity, and Monte Carlo techniques to solve radiation–conduction coupling in industrial furnaces, supporting the broader trend toward versatile, accurate, and scalable thermal simulation strategies.

Continuous-annealing furnaces are composed of multiple temperature-controlled zones. The tubular burners inside the heating section of the furnace are switched on or off to regulate the air temperature according to the desired set value. Often the zone air temperatures were considered as the primary factor affecting the heat transfer in the furnace. However, controlling the heat transfer process based on a single temperature data could sacrifice the process quality. This study numerically investigates the effect of positioning the active tubular burners to different zones of the furnace on the zone temperatures and on the heat transfer to glassware. In this regard a detailed numerical thermal model of an actual industrial-scale continuous container glass annealing furnace is developed. The details of the numerical model are first explained, and the validity of the model was assessed using temperature data acquired from actual furnace operation. Finally, the results of the cases with different active burner positions were discussed.

2 Numerical Methodology

2.1 Computational Domain and Boundary Conditions

The annealing furnace studied in this work is an industrial-scale continuous annealing furnace used in container glass production. The furnace features a linear conveyor system that transports glass containers through a series of thermally controlled zones. An isometric view of the furnace is given in Figure 1. The furnace has a double wall structure which forms a separate layer around the inner furnace region. The air sucked by the circulation fans mounted on top of the furnace is blown into this layer and fed back into the inner furnace region at the bottom. Thereby, the temperature uniformity in the zones is increased and the heat transfer between the furnace atmosphere and glassware is enhanced. The furnace consists of a total of nine thermal zones, categorized as five heating zones, four cooling zones. Heating zones are equipped with tubular radiant burners positioned at the top of the furnace. They are visible as green colored tubes in Figure 1. In normal operation, the burners in the first two zones usually operate continuously to heat up the glassware entering the furnace. The burners in the downstream zones usually operate in an intermittent manner and only burn a fraction of the gas that the burners in the first and second zones do. For this reason, for the validation of the numerical model only the burners in the first zone are activated. For the activated burners the tube temperatures are set to $677\ ^{o}C$ and the tube temperature of the remaining inactive burners are set to $527\ ^{o}C$. The tube temperature for the active burner is taken from the visuals of the burner tubes from the furnace inlet and from the infrared temperature measurements with a hand-held measurement system. The cooling zones in the last two zones are equipped with air vents where cooling air is sucked into the furnace to keep the desired temperature levels in the cooling zones. An exhaust opening is located upstream of these zones to remove hot exhaust gases from the furnace environment. The positions of the air vents and exhaust pipe are shown in Figure 2.

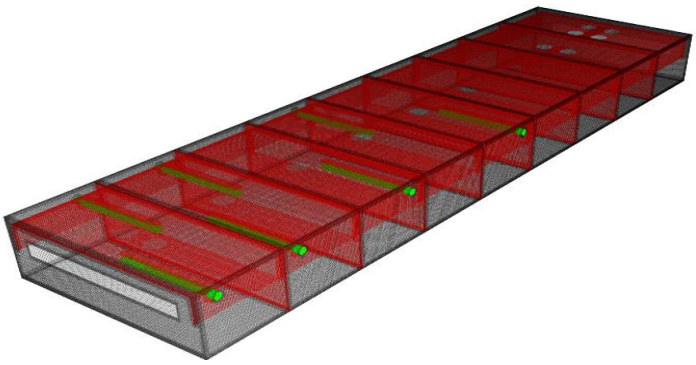


Figure 1: Isometric view of Furnace Geometry

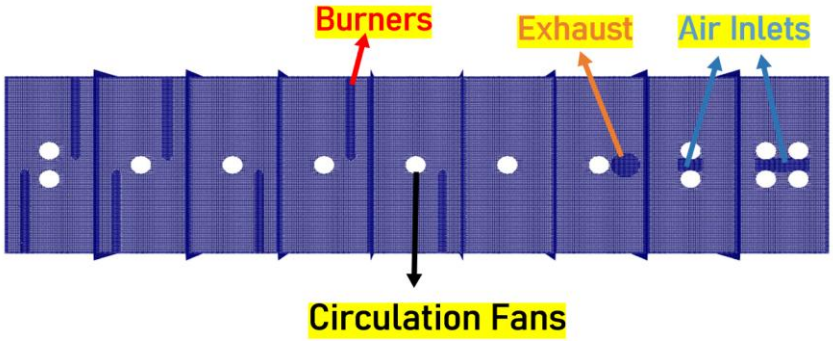


Figure 2: Inner Furnace Walls Mesh

At full capacity, the furnace holds 203 rows of glassware, with 40 glassware per row. the model includes over 8,000 individual glass objects. Glassware enters the furnace at a uniform initial temperature of 477 °C where they are heated up to the annealing temperature of glass that is around 520 °C then cooled back down in a controlled manner. The operating conditions of the furnace, including burners' wall temperatures, airflow rates, and conveyor speed, are imposed as boundary conditions. These values are derived from actual measurements at the plant and serve as critical inputs for the numerical simulation. These values are listed in

Table **1**.

Table 1. Furnace Geometric Parameters and Boundary Conditions used for the study

Parameter	Value
Total Furnace Length	20.25 m
Furnace Width	5.2 m
Furnace Height	2.0 m
Number of Zones	9 (5 Heating, 4 Cooling)
Number of Glassware Rows	203
Number of Glassware per Row	40
Total Number of Glassware	8,120
Glassware Temperature at Inlet	477 °C
Burner Tube Wall Temperature (Zone 1 – active)	677 °C
Burner Tube Wall Temperature (Zone 2-9 – inactive)	527 °C
Conveyor Speed	60 cm/min
Total Cooling Air Inlet Mass Flow Rate	2.6 kg/s
Cooling Air Temperatures	27 °C

2.2 Numerical Solvers and Models

The heat treatment process inside the continuous annealing furnace is modelled as a conjugate heat transfer problem. The modeling framework includes both the solid domain (glassware) and the fluid domain (furnace atmosphere), each governed by different equations and boundary conditions. The solid and fluid domains are solved by different solvers which work in an iterative manner. A transient heat conduction solver is used to solve the transient heat conduction in the solid domain that is composed of moving glassware. The moving solid domain consists of all the 40 glassware in the same row. Each glassware is composed of 500 hexahedral type volume cells and around 1200 triangular surface cells at the surface. Radiative heat transfer between the glassware and the furnace walls including the tubular burners is solved using an inhouse developed Monte Carlo ray tracing-based surface-to-surface radiation solver. In here, all the surfaces including the glassware are treated as black. Please consider that glass is a highly absorbing media for infrared radiation and black body assumption does not stay very far away from reality. Please refer to Altun and Basol [3] for more details of this solver. Convective heat transfer between the glassware and furnace atmosphere is also considered by the transient heat conduction solver. For this solver the furnace wall temperatures and furnace air atmosphere temperature distributions required for the convective heat transfer calculations are taken from the solution of the fluid domain. The convection in the furnace atmosphere is treated by a different solver. In this regard a steady state RANS solver "buoyantSimpleFoam" in OpenFoam is used. The solver utilizes SIMPLE algorithm for the solution of the RANS equations. The turbulence is modeled using k-ω turbulence model. Fluid domain mesh is shown in Figure 2 and 3. It consists of approximately 4 million hexahedral cells. It is important to point out that the glassware in the furnace are not directly resolved in this domain but modeled using porous zone approach. This approach dramatically decreased the mesh size required for the computation. In the same figure burners, circulation fans (white circles), exhaust pipe outlet and air vents are seen. At the air vent inlets found in the last two zones total pressure is set to the atmospheric pressure. The air flowrate into the furnace is regulated via the porosity value of the zones positioned at the air vents. At the exhaust static pressure with a negative gauge pressure is set to model the suction effect of the exhaust fan. The radiative heat transfer between the furnace walls is treated using the surface-to-surface radiation solver. The glassware surface temperatures are taken from the transient solid domain solver and used in the calculation of the radiative transfer between glassware and the furnace walls. For the calculation of the radiative heat transfer between glassware and the furnace walls the glassware geometry is used in full geometric detail requiring over 10 million surface cells for the 8120 glassware present inside the furnace. Convective heat transfer between the furnace atmosphere and the glassware is calculated by assuming a constant heat transfer coefficient on the glassware.

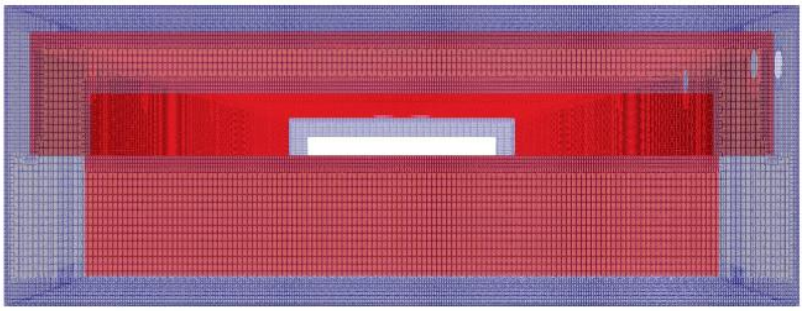


Figure 3: Double layer furnace geometry mesh viewed from the Inlet (RED – Inner Furnace Regions and BLUE – Outer Furnace Region)

2.3 Solver Coupling Strategy

The simulation framework employs two independent solvers that are coupled iteratively:

- A transient solver for the glass domain to resolve internal conduction over time as the glass moves through the furnace,
- A steady-state solver for the furnace air domain to compute flow and temperature fields as well as radiative heat exchange.

The coupling of these two solvers is carried out by transferring the temperature boundary conditions between the domains. In this regard, the transient conduction solver utilizes the furnace wall temperatures and air temperatures as boundary conditions to calculate the radiative and convective heat fluxes on the glassware. Next, the surface temperatures of the glassware are extracted from the solid domain and transferred to the steady RANS solver as boundary conditions where the convection and radiation inside the furnace atmosphere is solved. This process continues iteratively until convergence in both solvers is achieved. Additionally, the convergence of the steady state RANS solver is also monitored using temperature data at the zone midpoints. Figure 4 shows that about 3000 iterations are required for the temperature field to reach steady state. It is important to point out that the radiation model within the flow solver is called once every 10 iterations of the RANS solver.

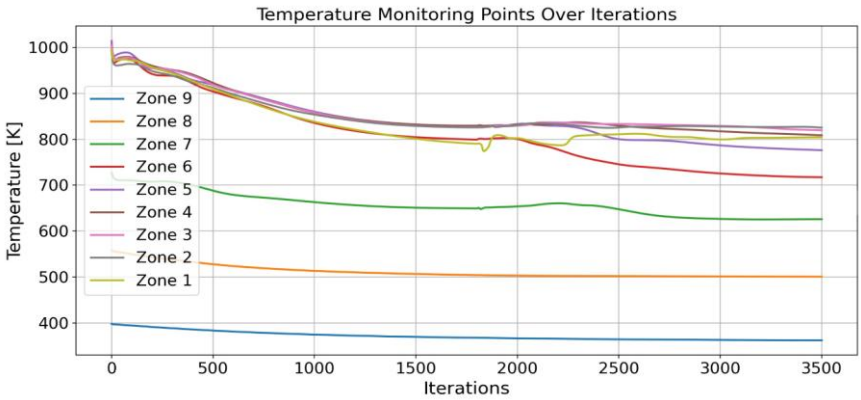


Figure 4: Convergence check by midpoint monitoring of temperatures on each zone.

2.4 Model Validation

The model is validated with real furnace data, and the convergence is checked by using temperatures at midpoints of each zone. As seen in Figure 5, zone temperatures close to the inlet and outlet of the furnace agree with the data. On the other hand, zone temperatures at the inner zones of the furnace are overpredicted by the model. In the real furnace data, the zone temperatures from the 3rd zone starts to decline even though the zones up until the fifth zone are listed as heating zones which are equipped with tubular burners. The decline in temperature from the 3rd zone onwards lets us consider cold air flow penetration from the furnace outlet towards the furnace inlet which is not considered in the existing model due to neglecting air outflow from the furnace inlet.

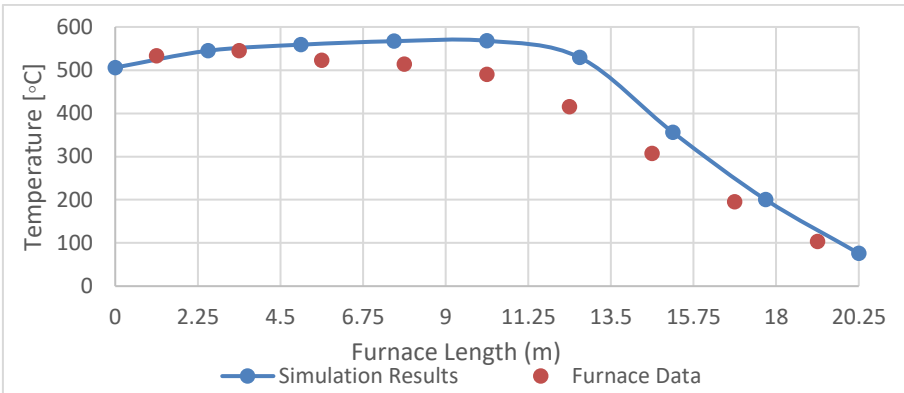


Figure 5: Air temperature prediction along the furnace by the model and Furnace Zone Temperature Data

3 Results

The effect of the active burner position on the annealing process is next investigated. Two cases are considered where the heaters in zone 1 are activated for Case 1 and the heaters in zone 2 are activated for Case 2. In both cases the burners only in a single zone are activated, all the others are deactivated. Resulting zone temperatures are shown in Figure 6 for Case 1 and Case 2. As shown, there are minor temperature differences between the cases. In Case 1, active burners in Zone 1 increase the zone temperature especially at the first half of the zone. In Case 2, the active burners in zone 2 results in elevated temperatures in zone 2 compared to zone 1. However, in both cases the zone temperature downstream of zone 2 are almost the same except the small difference in the zone temperatures in the cooling zones.

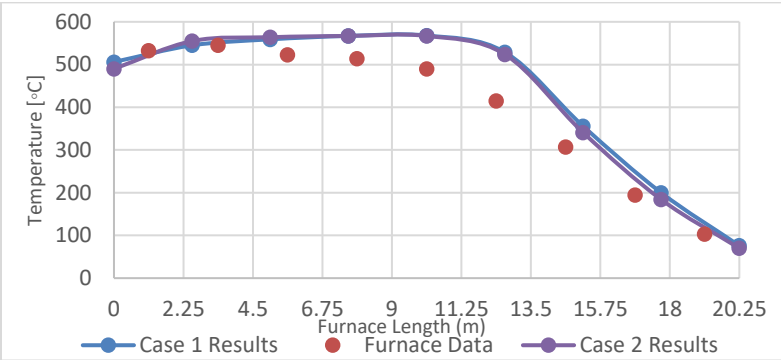


Figure 6: Air temperature prediction along the furnace by the model for Case 1, Case 2 and Furnace Zone Temperature Data

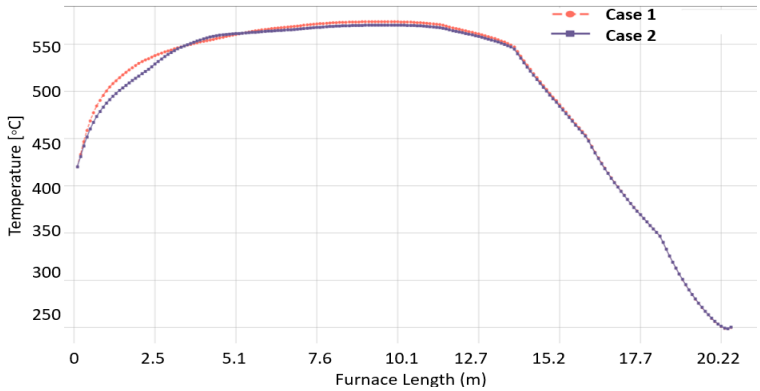
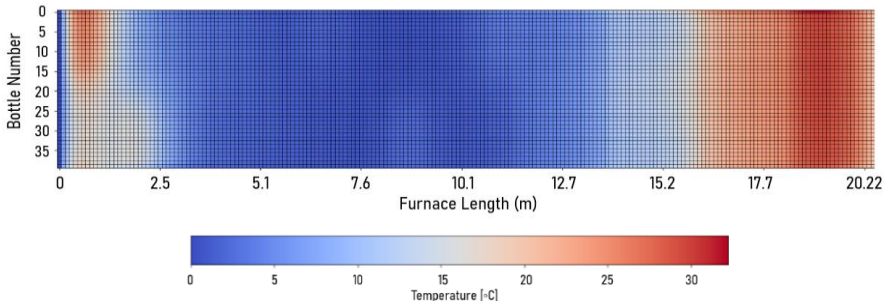


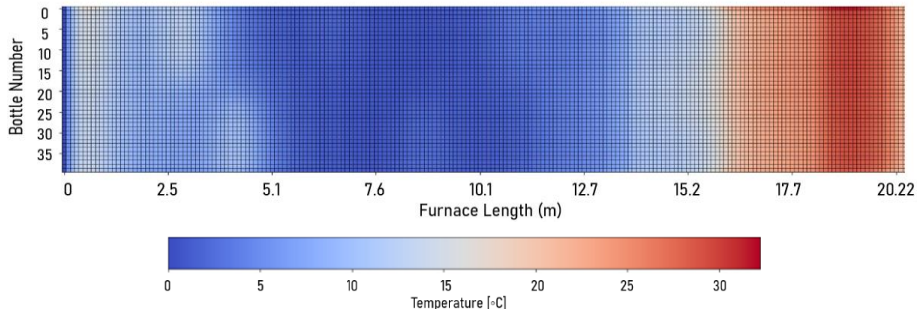
Figure 7: Row wise averaged glassware temperature along the furnace

Figure 7 shows the row-wise averaged glass temperatures throughout the furnace. Here, a clearer effect of changing the active burner position could be seen. In both cases glass temperatures enter the furnace at the same temperature. However, glass

temperatures show a steeper temperature rise in Case 1 due to them facing the active burners immediately after they enter the furnace. However, glassware in Case 2 is not subject to the radiative heating of the active burners at the furnace inlet, that is why they show a less intense heating in the first zone. However, when the glassware enters Zone 2 the active burners increase the temperature of glassware to the level of the glassware in Case 1. The row averaged glass temperatures are very similar to each other from the first two zones onwards.



(a) Case 1



(b) Case 2

Figure 8: Standard Deviations in Glassware temperature in the furnace (Case 1 vs. Case 2)

Figure 8 visualizes the standard deviation in glass temperatures, indicating the temperature nonuniformity inside the glass in the heating and cooling zones. This quantity is used as a measure of process quality. As shown, standard deviations in the first zone are higher in Case 1 as compared to Case 2. On the other hand, standard deviations in Zone 2 are higher in Case 2 compared to Case 1, also for Case 1 higher standard deviations were observed when comparing active heater zones

4 Conclusion

In this study, the annealing process in an industrial-scale continuous container glassware annealing furnace was numerically studied. The developed thermal model considered all heat transfer modes present in the furnace. Radiation heat transfer between the glassware and the furnace, as well as among glassware units, was solved using full-geometrical detail. For model validation, the predicted air temperature distribution along the furnace was compared with zone temperature data. At the inlet and outlet, predictions matched zone measurements, but temperatures midway through the furnace were over-predicted by about 50 °C to 100 °C. Next, the effect of changing active burner positions on zone and glass temperatures was studied. Results show that while burner position has a mild effect on zone temperatures, it significantly impacts the glassware temperatures. This is due to the dominance of radiative heating from burner tube walls. Shifting the active burners from Zone 1 to Zone 2 leads to more gradual and uniform heating of the glassware. Apart from the difference in heating patterns in the first two zones, no major changes in zone or glass temperatures were observed elsewhere in the furnace. This study highlights the importance of burner operational strategy in controlling heat transfer within the furnace.

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